



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT3042
Course title (English)	Honours Linear Algebra II
Course title (Chinese)	线性代数荣誉课程 (二)
Units	3
Language of Instruction	English
Description (English)	This is a continuation of MAT2042 (Honours Linear Algebra). Topics include: Vector space over fields: linear transformations and its matrix representations, basic category theory, quotient space and dual space, isomorphism theorems, tensor products, multilinear maps and the determinant. Modules: free modules and Noetherian modules, modules over principal ideal domains, Smith Normal Form. Operators on inner product spaces: characteristic and minimal polynomials, Jordan Normal Form, Schur's Lemma, spectral theorem of normal operators, polar decomposition
Description (Chinese)	本科目是 MAT2042 线性代数荣誉课程的延续。主题包括：域上的向量空间：线性变换及其矩阵表示、基本范畴论、商空间和对偶空间、同构定理、张量积、多线性映射和行列式。模：自由模和诺特模，主理想域上的模，史密斯范式。内积空间上的运算元：特征多项式和最小多项式、乔丹范式、舒尔引理、正规运算元的谱定理、极分解

2. Prerequisites / Co-requisites

A. Prerequisites

- MATH1002/MAT1012 (Calculus II)
- MATH2042 Linear Algebra, or A grade in MAT2040/MAT2041

B. Co-requisites

None.



3. Learning Outcomes

Knowledge:

- a) Students will understand vector spaces over any arbitrary field.
- b) Students will understand basic category theory and how they are applied to studying various construction of vector spaces.
- c) Students will learn the definition of determinants using tensor product and exterior products.
- d) Students will learn modules over a ring, and how the theorems in MAT2040/41/42 can be generalized to modules over principal ideal domains.
- e) Students will get to know the spectral theorem and polar decomposition theorem for normal operators over complex inner product spaces.

Skills

Generic:

- f) Students will be able to understand and write rigorous proofs.
- g) Students will be able to appreciate the close relations among various branches of mathematics.
- h) Students will be able to study higher-level mathematics and physics, and/or to continue their postgraduate studies in science or engineering.

Valued/Attitude:

- i) Students will appreciate the use of the abstract mathematical language in proving important theorems.
- j) Students will be able to understand the relations of linear algebra to different fields in mathematics, such as topology, differential geometry, mathematical physics and differential equations.
- k) Students will obtain the mathematical maturity to appreciate abstract concepts in theoretical science.

4. Course syllabus

Linear transformations of vector spaces and its matrix representations, basic category theory, quotient space and dual space, isomorphism theorems, tensor products, multilinear maps and the determinant.

Free modules and Noetherian modules, modules over principal ideal domains, Smith Normal Form.

Characteristic polynomials and minimal polynomials, Jordan Normal Form, Schur's Lemma, spectral theorem of normal operators of inner product space, polar



decomposition

5. Assessment Scheme

Component/ method	% weight
Assignments	10
Mid-term exam	35
Final exam	55

6. Grade descriptor

General

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to apply the principles in solving problems in novel situations, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study or research. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to familiar and standard situations in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity.
B	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
C	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
D	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials Has the ability to state the knowledge or application in simple terms.
F	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.



7. Feedback for evaluation

- A course questionnaire answered by every student after finishing all the lectures
- Reflections from teachers of more advanced courses
- Feedback from informal conversation over in the instructor's office and e-mail correspondence with students after class.

8. Reading

Required Reading

1. Advanced Linear Algebra, Steven Roman, 3RD EDITION, United States, Graduate Texts in Mathematics 135

9. Course components

Activity	Hours/week
Lectures	3
Readings & Assignments	6
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
1	Vector spaces and linear transformations
2	Matrix representations of linear transformations, similar matrices
3	Basic construction of vector spaces – direct sum, quotient space, dual space
4	Basic category theory, universal properties
5	Tensor product, Exterior product, the determinant
6	Modules over a ring, free modules
7	Quotient Modules, the correspondence theorem and isomorphism theorems
8	Ascending and descending chain condition, Noetherian ring and Noetherian modules
9	Modules over principal ideal domain
10	Primary decomposition theorem, Smith normal form
11	Eigenvalues and eigenvectors of linear operator, Jordan normal form
12	Inner product space, adjoint operators, Riesz representation theorem
13	Normal operators, spectral theorem and polar decomposition



11. **Any other information**

Prohibit the use of AI tools: AI tools (particularly GPT and generative AI tools) are not allowed in submitted reports. Any student who violates this rule will be considered to be engaging in academic dishonesty. The instructor may specify which AI tools are prohibited in this course.

12. **Version date**

Version number	1
As of (date)	2024/08/12